Statistics – Scenario-Based Questions

Part 1: Worked examples with calculations

Scenario 1: Delivery Time Analysis for an E-commerce Company

An e-commerce company tracks the delivery time (in minutes) of 15 orders:

25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95

The company wants to analyse delivery performance using percentiles and check for any unusual delivery times.

Question 1: Calculate Q1 and Q3

- **Q1 (25th percentile)** – 25% of the data lies below this value.
- **Q3 (75th percentile)** – 75% of the data lies below this value.

  Position of Q1 = 0.25 × (n + 1) = 0.25 × 16 = 4
  Position of Q3 = 0.75 × (n + 1) = 0.75 × 16 = 12

- Q1 = 4th value = **40**
- Q3 = 12th value = **80**

Question 2: Find the IQR

The interquartile range (IQR) measures the spread of the middle 50% of the data.

IQR = Q3 − Q1 = 80 − 40 = ***40***

Question 3: Detect outliers

Outliers are extreme values that fall outside these limits:

Lower bound = Q1 − 1.5 × IQR = 40 − 1.5 × 40 = −20
Upper bound = Q3 + 1.5 × IQR = 80 + 1.5 × 40 = 140

All delivery times lie between 25 and 95, which is inside the range −20 to 140. **Conclusion:** there are no outliers in this dataset.

Scenario 2: Mathematics Scores

A teacher is analysing the mathematics scores of the students in her class:

45, 50, 55, 60, 60, 62, 63, 65, 90, 95

Questions

- Calculate the mean, median and mode of the scores.
- Explain why the median may be a better representation than the mean here.

Answer:

- **Mean** = (45 + 50 + 55 + 60 + 60 + 62 + 63 + 65 + 90 + 95) ÷ 10 = 645 ÷ 10 = **64.5**
- **Median** = average of the 5th and 6th values = (60 + 62) ÷ 2 = **61**
- **Mode** = **60** (it appears twice)

Explanation

Most students scored between 45 and 65, but two high scores (90 and 95) pull the mean up to 64.5. The median (61) is not affected by these extreme values, so it gives a better picture of a typical student's performance. Since the mean is greater than the median, the data is slightly right-skewed.

Scenario 3: Daily Customer Visits

A grocery store manager records the number of customers visiting the store each day:

5, 10, 8, 15, 20, 5, 12, 14, 10, 18

Question: Create a frequency distribution table for this data.

Number of Customers	Frequency
5	2
8	1
10	2
12	1
14	1
15	1
18	1
20	1
Total	10

Scenario 4: Multicollinearity in a Real Estate Model

A real estate price model uses three input variables: **house size**, **number of rooms** and **number of bathrooms**.

Question: How can you detect multicollinearity?

Answer:

- Calculate the **Variance Inflation Factor (VIF)** for each variable.
- A VIF greater than 10 (some analysts use 5) indicates strong multicollinearity.
- A high VIF means the variables are highly correlated with each other, which makes the model's coefficients unstable and harder to interpret.

Scenario 5: Testing a Blood Pressure Medicine

A company developed a new medicine to lower blood pressure. One group received the medicine and another group received a placebo.

Question: How can the company check whether the new medicine works?

Answer:

- **Set up the hypotheses:**
  - **H_0:** The medicine does not lower blood pressure.
  - **H_1:** The medicine lowers blood pressure.
- **Perform a t-test** and find the p-value (the probability of seeing this result if the medicine actually had no effect).
- **Decision:** if the p-value is less than 0.05, reject H_0 – the medicine is likely effective.

Scenario 6: Identifying Outliers in Sales Data

A company wants to find any unusual spikes in its sales.

Question: How can the company detect outliers in its sales data?

Answer:

- **Step 1:** Calculate Q1, Q3 and the interquartile range (IQR = Q3 − Q1).
- **Step 2:** Mark a value as an outlier if it is below Q1 − 1.5 × IQR or above Q3 + 1.5 × IQR.

Scenario 7: Understanding Customer Satisfaction

A restaurant surveyed its customers and recorded satisfaction on a scale of 1 to 5:

5, 4, 4, 5, 3, 4, 5, 2, 4, 3

Question: How can the restaurant summarise overall satisfaction?

Answer:

- **Mode = 4** – the most common rating.
- **Mean = 39 ÷ 10 = 3.9** and **Median = 4** – both confirm a typical rating of about 4.
- 7 out of 10 customers gave a rating of 4 or 5, so overall satisfaction is high.